



Voice-Enabled Task Tracker - API Documentation

Base URL

```
http://localhost:4000/api
```

Authentication

Currently **no authentication required** (single-user app).



Task Management Endpoints

1. Get All Tasks

Request

```
GET /tasks?status=todo&priority=high&search=review&dueFrom=2025-12-06&dueTo=2025-12-15
```

Query Parameters

Parameter	Type	Required	Description	Example
status	string	No	Filter by status: <code>todo</code> , <code>in_progress</code> , <code>done</code>	?status=in_progress
priority	string	No	Filter by priority: <code>low</code> , <code>medium</code> , <code>high</code> , <code>critical</code>	?priority=high
search	string	No	Search by title or description (case-insensitive)	?search=bug
dueFrom	ISO date	No	Filter tasks due from this date (inclusive)	?dueFrom=2025-12-06
dueTo	ISO date	No	Filter tasks due until this date (inclusive)	?dueTo=2025-12-15

Response

Success (200 OK)

```
[
  {
    "id": "550e8400-e29b-41d4-a716-446655440000",
    "title": "Review pull request",
    "description": "Check authentication module changes",
    "status": "todo",
    "priority": "high",
    "due_date": "2025-12-06T18:00:00.000Z",
    "created_at": "2025-12-05T10:30:00.000Z",
    "updated_at": "2025-12-05T10:30:00.000Z"
  },
  {
    "id": "550e8400-e29b-41d4-a716-446655440001",
    "title": "Fix login bug",
    "description": null,
    "status": "in_progress",
    "priority": "critical",
    "due_date": "2025-12-06T09:00:00.000Z",
    "created_at": "2025-12-04T14:20:00.000Z",
    "updated_at": "2025-12-05T08:15:00.000Z"
  }
]
```

Error Responses

Bad Request (400)

```
{
  "message": "Invalid filter parameter"
}
```

Server Error (500)

```
{
  "message": "Database connection failed"
}
```

Example Requests

```
# Get all tasks
curl "http://localhost:4000/api/tasks"

# Get only urgent tasks
curl "http://localhost:4000/api/tasks?priority=critical"

# Get in-progress tasks due this week
```

```
curl "http://localhost:4000/api/tasks?status=in_progress&dueFrom=2025-12-06&dueTo=2025-12-12"

# Search for "code review"
curl "http://localhost:4000/api/tasks?search=code%20review"

# Combine filters
curl "http://localhost:4000/api/tasks?status=todo&priority=high&search=urgent"
```

2. Get Single Task

Request

```
GET /tasks/:id
```

Path Parameters

Parameter	Type	Required	Description
id	UUID	Yes	Task ID

Response

Success (200 OK)

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "title": "Review pull request",
  "description": "Check authentication module changes",
  "status": "todo",
  "priority": "high",
  "due_date": "2025-12-06T18:00:00.000Z",
  "created_at": "2025-12-05T10:30:00.000Z",
  "updated_at": "2025-12-05T10:30:00.000Z"
}
```

Not Found (404)

```
{
  "message": "Task not found"
}
```

Example Request

```
curl "http://localhost:4000/api/tasks/550e8400-e29b-41d4-a716-446655440000"
```

3. Create Task

Request

```
POST /tasks
Content-Type: application/json
```

Body Parameters

Field	Type	Required	Constraints	Example
title	string	Yes	1-255 chars, non-empty	"Review PR"
description	string	No	0-5000 chars	"Check auth changes"
priority	string	No	low, medium, high, critical	"high"
status	string	No	todo, in_progress, done	"todo"
dueDate	ISO datetime	No	Valid ISO 8601 format	"2025-12-06T18:00:00Z"

Response

Created (201)

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "title": "Review pull request",
  "description": "Check authentication module changes",
  "status": "todo",
  "priority": "high",
  "due_date": "2025-12-06T18:00:00.000Z",
  "created_at": "2025-12-05T10:30:00.000Z",
  "updated_at": "2025-12-05T10:30:00.000Z"
}
```

Bad Request (400)

```
{
  "message": "Title is required"
}
```

Example Requests

```
# Minimal task
curl -X POST "http://localhost:4000/api/tasks" \
  -H "Content-Type: application/json" \
  -d '{
    "title": "Fix login bug"
  }'

# Complete task
curl -X POST "http://localhost:4000/api/tasks" \
  -H "Content-Type: application/json" \
  -d '{
    "title": "Review pull request",
    "description": "Check authentication module changes",
    "priority": "high",
    "status": "todo",
    "dueDate": "2025-12-06T18:00:00Z"
  }'
```

4. Update Task

Request

```
PUT /tasks/:id
Content-Type: application/json
```

Path Parameters

Parameter	Type	Required	Description
id	UUID	Yes	Task ID

Body Parameters (All Optional)

Field	Type	Constraints	Example
title	string	1-255 chars	"Updated title"
description	string	0-5000 chars	"New description"
priority	string	low, medium, high, critical	"critical"
status	string	todo, in_progress, done	"in_progress"
dueDate	ISO datetime	Valid ISO 8601 or null	"2025-12-07T09:00:00Z"

Response

Success (200)

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "title": "Updated title",
  "description": "Updated description",
  "status": "in_progress",
  "priority": "critical",
  "due_date": "2025-12-07T09:00:00.000Z",
  "created_at": "2025-12-05T10:30:00.000Z",
  "updated_at": "2025-12-05T11:45:00.000Z"
}
```

Not Found (404)

```
{
  "message": "Task not found"
}
```

Bad Request (400)

```
{
  "message": "Invalid status"
}
```

Example Requests

```
# Update single field
curl -X PUT "http://localhost:4000/api/tasks/550e8400-e29b-41d4-a716-446655440000" \
  -H "Content-Type: application/json" \
  -d '{
    "status": "in_progress"
  }'

# Update multiple fields
curl -X PUT "http://localhost:4000/api/tasks/550e8400-e29b-41d4-a716-446655440000" \
  -H "Content-Type: application/json" \
  -d '{
    "title": "New title",
    "priority": "critical",
    "status": "done",
    "dueDate": "2025-12-10T18:00:00Z"
  }'
```

```
# Clear due date
curl -X PUT "http://localhost:4000/api/tasks/550e8400-e29b-41d4-a716-446655440000" \
  -H "Content-Type: application/json" \
  -d '{
    "dueDate": null
  }'
```

5. Delete Task

Request

```
DELETE /tasks/:id
```

Path Parameters

Parameter	Type	Required	Description
<code>id</code>	UUID	Yes	Task ID

Response

Success (200)

```
{
  "message": "Task deleted successfully"
}
```

Not Found (404)

```
{
  "message": "Task not found"
}
```

Example Request

```
curl -X DELETE "http://localhost:4000/api/tasks/550e8400-e29b-41d4-a716-446655440000"
```

6. Parse Voice Transcript

Request

```
POST /voice/parse
Content-Type: application/json
```

Body Parameters

Field	Type	Required	Description	Example
transcript	string	Yes	Raw voice-to-text transcript	"Create urgent task by Friday"

Response

Success (200)

```
{
  "transcript": "Create urgent task to review code by Friday morning",
  "parsed": {
    "title": "review code",
    "priority": "high",
    "status": "todo",
    "dueDate": "2025-12-12T09:00:00.000Z"
  }
}
```

Bad Request (400)

```
{
  "message": "transcript is required"
}
```

Server Error (500)

```
{
  "message": "Failed to parse transcript with AI"
}
```

Supported Formats

Priority Keywords:

- low, low priority → low

- `medium, normal` → `medium`
- `high, urgent, important` → `high`
- `critical, very high, asap` → `critical`

Status Keywords:

- `todo, to do` → `todo`
- `in progress, doing` → `in_progress`
- `done, completed, finished` → `done`

Date Phrases:

- `tomorrow` → next day
- `day after tomorrow` → +2 days
- `next Monday, next Friday` → upcoming weekday
- `in 3 days, in 2 weeks` → relative offset
- `January 15, 15th Jan` → absolute date
- `morning (9 AM), afternoon (3 PM), evening (6 PM), night (9 PM)`

Example Requests

```
# Simple task
curl -X POST "http://localhost:4000/api/voice/parse" \
  -H "Content-Type: application/json" \
  -d '{
    "transcript": "Create task to review code"
  }'

# Complex voice input
curl -X POST "http://localhost:4000/api/voice/parse" \
  -H "Content-Type: application/json" \
  -d '{
    "transcript": "Urgent: fix the login bug by next Friday morning, critical
priority"
  }'

# With relative date
curl -X POST "http://localhost:4000/api/voice/parse" \
  -H "Content-Type: application/json" \
  -d '{
    "transcript": "Low priority: update documentation in 2 weeks"
  }'
```

7. Create Task from Voice (Optional)

Request

```
POST /voice/create-from-voice
Content-Type: application/json
```

Body Parameters

Field	Type	Required	Description
transcript	string	Yes	Raw voice transcript

Response

Created (201)

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "title": "review code",
  "description": null,
  "status": "todo",
  "priority": "high",
  "due_date": "2025-12-12T09:00:00.000Z",
  "created_at": "2025-12-05T10:30:00.000Z",
  "updated_at": "2025-12-05T10:30:00.000Z"
}
```

Example Request

```
curl -X POST "http://localhost:4000/api/voice/create-from-voice" \
-H "Content-Type: application/json" \
-d '{
  "transcript": "Create high priority task to review code by Friday"
}'
```



Data Formats

Status Values

- "todo" → To Do
- "in_progress" → In Progress
- "done" → Done

Priority Values

- "low" → Low Priority
- "medium" → Medium Priority
- "high" → High Priority
- "critical" → Critical/Urgent

Date Format

ISO 8601: 2025-12-06T18:00:00.000Z

HTTP Status Codes

Code	Meaning	Use Case
200	OK	Successful GET, PUT, DELETE
201	Created	Task successfully created (POST)
400	Bad Request	Invalid input, missing required field
404	Not Found	Task ID doesn't exist
500	Server Error	Database or parsing error

Sample API Flows

Flow 1: Create Task Manually

```
POST /tasks
{
  "title": "Review PR",
  "priority": "high",
  "dueDate": "2025-12-06T18:00:00Z"
}
→ 201 Created
```

Flow 2: Voice to Task

```
POST /voice/parse
{
  "transcript": "urgent task by Friday"
}
→ 200 OK with parsed data
```

```
POST /tasks
{...parsed data...}
→ 201 Created
```

Flow 3: Update Task Status

```
PUT /tasks/:id
{
  "status": "in_progress"
}
→ 200 OK
```

Flow 4: Delete Task

```
DELETE /tasks/:id
→ 200 OK
```

Error Handling

All errors return JSON with a `message` field:

```
{
  "message": "Error description here"
}
```

Common Errors:

Error	Cause	Fix
Title is required	Empty/missing title	Add <code>title</code> field
Invalid status	Wrong status value	Use <code>todo</code> , <code>in_progress</code> , <code>done</code>
Invalid priority	Wrong priority value	Use <code>low</code> , <code>medium</code> , <code>high</code> , <code>critical</code>
Invalid dueDate	Malformed date	Use ISO 8601 format
Task not found	ID doesn't exist	Check task ID
Failed to parse transcript	AI error	Retry or use manual form